

Single-Set Quantitative Selection Principle

This note implements the concrete selection-principle step from `plan1.md`. The purpose is not to replace the whole Brasco–De Philippis–Velichkov proof, but to isolate the part that can be made quantitative without changing their free-boundary machinery.

The main output is a single-set form of BDV Proposition 4.4. Starting from one near-counterexample, it constructs a selected quasi-open minimizer whose energy-to-asymmetry ratio is controlled up to constants.

1 Normalization

Use the BDV torsion convention

$$E(\Omega) = \min_{u \in H_0^1(\Omega)} \left\{ \frac{1}{2} \int_{\Omega} |\nabla u|^2 - \int_{\Omega} u \right\} = -\frac{1}{2} \int_{\Omega} u_{\Omega}.$$

Fix

$$|\Omega| = |B_1| = \omega_N.$$

Then the Saint-Venant deficit is

$$\delta(\Omega) := E(\Omega) - E(B_1) \geq 0.$$

For the selection step, use BDV's smoother asymmetry

$$\alpha(\Omega) := \int_{\Omega \Delta B_1(x_{\Omega})} |1 - |x - x_{\Omega}|| \, dx,$$

where x_{Ω} is the barycenter. BDV Lemma 4.2 gives, for bounded $\Omega \subset B_R$,

$$|\Omega \Delta B_1(x_{\Omega})|^2 \lesssim_N \alpha(\Omega),$$

$$|\alpha(\Omega_1) - \alpha(\Omega_2)| \lesssim_R |\Omega_1 \Delta \Omega_2|,$$

and, for nearly spherical sets,

$$\alpha(\Omega) \lesssim_N \|\varphi\|_{L^2(\partial B_1)}^2.$$

Thus the right badness ratio for the selection step is

$$Q_{\alpha}(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha(\Omega)}.$$

The final Fraenkel statement is recovered later from $\mathcal{A}(\Omega)^2 \lesssim \alpha(\Omega)$.

2 Penalized Problem

Let $\Omega_0 \subset B_R$, $R \geq 2$, satisfy

$$|\Omega_0| = |B_1|, \quad \varepsilon := \alpha(\Omega_0) > 0, \quad \delta := E(\Omega_0) - E(B_1), \quad q := \delta/\varepsilon.$$

Assume $q \ll 1$. Choose

$$\tau := q^{1/4},$$

or more generally any τ with

$$q \ll \tau^2 \ll 1.$$

Let $f_{\hat{\eta}}$ be the BDV volume penalty from Lemma 4.5:

$$f_{\hat{\eta}}(s) = \begin{cases} \hat{\eta}(s - \omega_N), & s \leq \omega_N, \\ (s - \omega_N)/\hat{\eta}, & s \geq \omega_N. \end{cases}$$

Set

$$F_{\hat{\eta}}(U) := E(U) + f_{\hat{\eta}}(|U|).$$

BDV Lemma 4.5 says that, for $\hat{\eta} = \hat{\eta}(N, R)$, B_1 is a minimizer of $F_{\hat{\eta}}$ among sets in B_R .

Now minimize over quasi-open $U \subset B_R$:

$$\mathcal{G}_{\varepsilon, \tau}(U) := F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}.$$

This is exactly BDV's penalized functional (4.10), with the sequence parameter σ replaced by the single-set parameter τ .

3 Quantitative Selected-Minimizer Estimates

Assume $\tau \leq \tau_0(N, R)$, small enough that the BDV existence and quasi-minimizer estimates hold with τ in place of σ . Let U_* be a minimizer of $\mathcal{G}_{\varepsilon, \tau}$.

Lemma 1. Energy and Asymmetry Preservation Before Normalization

The minimizer U_* satisfies

$$0 \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta,$$

and

$$|\alpha(U_*) - \varepsilon| \leq \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}.$$

In particular, if $q = \delta/\varepsilon \leq 1$ and $\tau = q^{1/4}$, then

$$|\alpha(U_*) - \alpha(\Omega_0)| \leq C\alpha(\Omega_0)q^{1/4}.$$

Proof. By minimality against Ω_0 ,

$$F_{\hat{\eta}}(U_*) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \leq F_{\hat{\eta}}(\Omega_0) + \varepsilon.$$

Since $|\Omega_0| = |B_1|$,

$$F_{\hat{\eta}}(\Omega_0) = E(\Omega_0) = F_{\hat{\eta}}(B_1) + \delta.$$

Because B_1 minimizes $F_{\hat{\eta}}$,

$$F_{\hat{\eta}}(U_*) \geq F_{\hat{\eta}}(B_1).$$

Therefore

$$F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta,$$

and

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \leq \varepsilon + \delta.$$

Squaring the last inequality gives

$$\tau^2(\alpha(U_*) - \varepsilon)^2 \leq 2\varepsilon\delta + \delta^2.$$

This proves the claim.

Lemma 2. Volume Error

There is $C = C(N, R)$ such that

$$||U_*| - |B_1|| \leq C\delta.$$

Proof. Let B_{r_*} be a ball with $|B_{r_*}| = |U_*|$. By rearrangement,

$$F_{\hat{\eta}}(B_{r_*}) \leq F_{\hat{\eta}}(U_*).$$

BDV Lemma 4.5 gives

$$F_{\hat{\eta}}(B_{r_*}) - F_{\hat{\eta}}(B_1) \geq \frac{|r_* - 1|}{C_4(N, R)}.$$

Together with Lemma 1,

$$|r_* - 1| \leq C\delta.$$

Since $|B_{r_*}| = \omega_N r_*^N$, this implies

$$||U_*| - |B_1|| \leq C\delta.$$

Lemma 3. Ratio Preservation After Volume Normalization

Let

$$\lambda := \left(\frac{|B_1|}{|U_*|} \right)^{1/N}, \quad \tilde{\Omega} := \lambda U_*.$$

After translating if needed, $\tilde{\Omega}$ has volume $|B_1|$, and

$$\alpha(\tilde{\Omega}) \geq \left(1 - Cq^{1/4}\right) \alpha(\Omega_0),$$

$$E(\tilde{\Omega}) - E(B_1) \leq C(E(\Omega_0) - E(B_1)),$$

provided $q \leq q_0(N, R)$ and $\alpha(\Omega_0)$ is in the small regime. Consequently,

$$Q_\alpha(\tilde{\Omega}) \leq CQ_\alpha(\Omega_0).$$

Proof sketch. Lemma 2 gives

$$|\lambda - 1| \leq C\delta = C\varepsilon q.$$

Since $q \ll 1$, this error is smaller than the asymmetry-preservation error $\varepsilon q^{1/4}$. The selected minimizer has the perimeter bound from the BDV existence argument, so the small dilation satisfies

$$|U_* \Delta \lambda U_*| \leq C(N, R)|\lambda - 1|.$$

The Lipschitz property of α on bounded sets then gives

$$|\alpha(\tilde{\Omega}) - \alpha(U_*)| \leq C\varepsilon q.$$

Combining with Lemma 1 gives

$$|\alpha(\tilde{\Omega}) - \varepsilon| \leq C\varepsilon q^{1/4}.$$

For the energy, use

$$E(\lambda U_*) = \lambda^{N+2} E(U_*),$$

the bound $F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta$, the Lipschitz control of $f_{\hat{\eta}}$ in the volume variable, and $|\lambda - 1| \leq C\delta$. Since $U_* \subset B_R$, the energy is bounded in terms of N, R , so the dilation changes the deficit by $O_{N,R}(\delta)$. Thus

$$E(\tilde{\Omega}) - E(B_1) \leq C\delta.$$

The ratio estimate follows from the two displayed bounds.

4 Free-Boundary Regularity Inherited from BDV

The minimizer $U_* = \{u_* > 0\}$ satisfies the perturbed one-phase minimality inequality

$$\frac{1}{2} \int |\nabla u_*|^2 - \int u_* + f_{\hat{\eta}}(|\{u_* > 0\}|) \leq \frac{1}{2} \int |\nabla v|^2 - \int v + f_{\hat{\eta}}(|\{v > 0\}|) + C_R \tau |\{u_* > 0\} \Delta \{v > 0\}|$$

for all $v \in H_0^1(B_R)$. This is BDV (4.19) with τ in place of σ .

Hence, for $\tau \leq \tau_0(N, R)$, the Alt-Caffarelli estimates used in BDV apply:

- nondegeneracy of u_* near the free boundary,
- Lipschitz control,
- density estimates for both phases,
- reduced-boundary structure,
- smoothness once the free boundary is sufficiently flat.

The Euler-Lagrange Bernoulli coefficient has the same structure as BDV Lemma 4.16:

$$q_{u_*}(x)^2 = \Lambda + \frac{\tau^2(\alpha(U_*) - \varepsilon)}{\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2}} \cdot \Psi_{U_*}(x),$$

where Ψ_{U_*} is the smooth α -variation term. The prefactor is bounded by τ , so q_{u_*} has uniform C^k bounds near ∂U_* , exactly as in BDV Lemma 4.16.

5 Quantitative Entry into Hausdorff Closeness

The compactness step in BDV can be partially quantified as follows.

Lemma 4. From α and Density to Hausdorff Distance

Assume $U \subset B_R$ satisfies the two-sided density estimates

$$|U \cap B_r(x)| \geq cr^N, \quad |B_r(x) \setminus U| \geq cr^N$$

for every $x \in \partial U$ and every $r \leq r_0$. If, after centering,

$$|U \Delta B_1| \leq M,$$

then

$$d_H(\partial U, \partial B_1) \leq C(N, R, c, r_0) M^{1/N}.$$

Using BDV Lemma 4.2,

$$M \lesssim_N \alpha(U)^{1/2},$$

so selected minimizers satisfy

$$d_H(\partial U, \partial B_1) \leq C \alpha(U)^{1/(2N)}.$$

Proof sketch. If $x \in \partial U$ and $\text{dist}(x, \partial B_1) = d$, then for $r \sim d$ the ball $B_r(x)$ lies either inside B_1 or outside B_1 . The density estimate forces a chunk of size $\gtrsim r^N$ in $U \Delta B_1$. Thus $d^N \lesssim M$.

Conversely, if $y \in \partial B_1$ is distance d from ∂U , then $B_d(y)$ is either contained in U or disjoint from U . Either way, a fixed fraction of $B_d(y)$ lies in $U \Delta B_1$, so again $d^N \lesssim M$.

6 Remaining Quantitative Gap

BDV's final regularity passage is

$$\begin{aligned} L^1 \text{ closeness} &\Rightarrow \text{Kuratowski/Hausdorff closeness} \\ &\Rightarrow \text{flatness at a fixed scale} \\ &\Rightarrow C^{1,\gamma} \text{ free-boundary graph} \Rightarrow C^\infty \text{ convergence.} \end{aligned}$$

Lemma 4 above supplies the Hausdorff step quantitatively for selected minimizers. The remaining task is to make the Alt-Caffarelli flatness threshold explicit enough to state

$$\alpha(U_*) \leq \varepsilon_0(N, R, \tau) \implies \partial U_* \text{ is a global } C^{1,\gamma} \text{ graph over } \partial B_1.$$

Once this graph entry is available, BDV's nearly spherical second variation gives

$$E(\tilde{\Omega}) - E(B_1) \gtrsim_N \|\varphi\|_{H^{1/2}(\partial B_1)}^2 \gtrsim_N \alpha(\tilde{\Omega}),$$

contradicting $Q_\alpha(\tilde{\Omega}) \ll 1$.

Thus the selection route has been reduced to one concrete missing quantitative regularity statement: track the flatness constants in the Alt-Caffarelli improvement step for the selected minimizers.